Essential in SketchUp (40 Hours)				
S.No	**Session**	**Topics Covered**	**Type**	**Duration (hrs)**
1	Session 1	**Introduction** Getting Started with SketchUp Understanding the Templates GUI Understanding SketchUp Interface Understanding Various Menu Understanding the Status Bar Understanding the Default Tray Understanding the Toolbars Customization Managing Trays Mouse Control & SketchUp preferences Creating Saving Opening a model Importing and Exporting Different Types file of SketchUp Axes Understanding the Axes in SketchUp Reorienting the Axes Inference Concept of Inference Understanding the Various inference types Locking the Inference	Theory/Demo	2.0 Hrs
2	Session 2	Practice 01: Importing Different Types of file into SketchUp	Practical	2.0 Hrs
3	Session 3	**Modelling Of Entities 1** Drawing Tool Understanding the line tool Constructing a surface Divide and Joining a face Understanding the various input method Rectangle understanding the rectangle tool and various dimensions inputs Drawing a Rotated rectangle entity. Understanding the creation of circle entity and its various parameters Ellipse Drawing an ellipse from circle Understanding the creation of Polygon entity and its various parameters Arc Drawing an Arc using various Arc tools 2 point Arc 3 point Arc Pie tool Drawing a freehand Shapes Selection of an object Understanding the Select tool selecting multiple entities Adding and subtracting from a selection set.	Theory/Demo	2.0 Hrs
4	Session 4	Practice 02: Create a simple 2D floor plan using the drawing tools	Practical	2.0 Hrs
5	Session 5	**Modelling Of Entities 2** Modify Tools Editing Shapes using Entity Info option Pushing and pulling shapes into 3D Cutting a 3D Shape out of a model Understanding the move tool Stretching geometry using Move tool Resizing Surface Entities Creating Copies using Move Tool and Rotate Tool Creating Multiple Copies/Array using Move tool and Rotate Tool offsetting Line from an existing geometry. Undo recent & Eraser tool	Theory/Demo	2.0 Hrs
6	Session 6	Practice 03: Create a 3D floor plan using modify tools.	Practical	2.0 Hrs
		Modelling Of Entities 3 Modify Tools		

Essential in SketchUp (40 Hours)				
7	Session 7	Understand the Dividing, Splitting and Exploding Entities Understand the Flipping, Mirroring and Rotating Scaling a model using Scale tool Stretching a model using scale tool Understand the Follow Me Tool Automatic and Manual Extrusion with Follow Me Tool Modelling Revolved Shape, Soften/ Smooth Edges of Geometry & Hiding Geometry Tape Measure Tool & Protractor Tool Measuring Distance/ Create guides Measuring an angle Editing Guidelines	Theory/Demo	2.0 Hrs
8	Session 8	Practice 04: Create Vase and Sun Shade using Follow Me	Practical	2.0 Hrs
9	Session 9	**Modelling Of Entities 4** Annotative Model Understanding the Annotation tools Adding Text Text as a Model Entity Dimension Tool Matching a photo to a model Match Photo's Colour- coded tools Creating a 3D Model from a photo Deleting a matched photo	Theory/Demo	2.0 Hrs
10	Session 10	Practice 05: Apply Text, Dimension to the model and Create building model using Matching photo.	Practical	2.0 Hrs
11	Session 11	**Groups** Grouping Geometry Working with hierarchies in the outliners Renaming groups and components Finding groups and components Identifying a group/Components Restructuring a model's hierarchy Controlling visibility with ;ayers.	Theory/Demo	2.0 Hrs
12	Session 12	Practice 06: Create Table in Group using Follow Me and Revolving tool.	Practical	2.0 Hrs
13	Session 13	**Components** Getting Started with Components Understanding the components Creating a component Editing a component Editing all instances of a component Making Dynamic component Adding attributes Defining Attribute Values and testing Inserting, replacing and Interacting with dynamic Component Searching for component Adjusting a component's inserting point Editing component.	Theory/Demo	2.0 Hrs
14	Session 14	Practice 07: Create Door in Components and Modify its properties.	Practical	2.0 Hrs
15	Session 15	**Presenting Your Model** Viewing a Model Camera Changing the view or focal length Section To place a section plane Filling voids in section cuts Creating new geometry from a section Showing/hiding section planes Exporting section cut effects Style Choosing a style Editing Styles Style Editing options Managing in-model styles and collections Sharing styles	Theory/Demo	2.0 Hrs

Essential in SketchUp (40 Hours)				
		Adding fog effects to model		
16	Session 16	Practice 08: Apply style and Create Scenes of SketchUp Model using Camera.	Practical	2.0 Hrs
17	Session 17	**Presenting Your Model** Applying Material & Texture Adding colour and Textures with material Applying Material & Texture Replacing one material with another Editing Material-Editing Option Calculating Material area Managing and organizing material Creating new material Applying Texture Adding Geographic location from Google Map Applying Real-time shadows.	Theory/Demo	2.0 Hrs
18	Session 18	Practice 09: Apply Material and texture to the model. Create new material to the model.	Practical	2.0 Hrs
19	Session 19	**3D Model to 2D** SketchUp Layout Introduction Template Selection Understanding the GUI Creating Documents in Layout Presenting layout Documents Scale Changing Views Changing Model Style Changing Dimension Style Inserting Text & Changing Text Style Exporting/ printing Layout Document	Theory/Demo	2.0 Hrs
20	Session 20	Practice 10: Convert your 3D SketchUp to 2D layout	Practical	2.0 Hrs